Revenue Workspace — Product Requirements v1

Status: proposed build specification for review. **Date:** 6 October 2026. **Working name:** Revenue Workspace. **Planning horizon:** two months on Azure, delivered in tested phases for the December 2026 Huawei demo. **Language:** English only. **Current authorization:** documentation only.

This PRD supersedes the earlier GTM-only, bilingual, standalone-demo and Huawei-first directions for future development. It covers the complete product roadmap: CRM, GTM, sales, marketing, campaigns, relationships, design assets, AI assistance, automation and reporting. The existing local demonstrator remains historical evidence of some workflow behaviors; it does not establish native Twenty, OpenDesign, provider or cloud readiness.

1. Product purpose and users

Build one AI-assisted revenue workspace on Twenty. A user can research an account, qualify a lead, prepare a campaign, generate branded material, conduct follow-up, manage a deal and report results without repeatedly reconstructing commercial context in separate tools.

The bottom-right assistant is a persistent working companion. It follows the user's current page and selection, explains records with their sources, navigates the application, and executes permitted actions. Its Design mode uses OpenDesign to produce and revise assets in the same conversation. Twenty supplies the CRM foundation and the product design system.

Initial users are agencies and consultancies managing B2B campaigns and client relationships, including a campaign/partnership lead such as Lara. The product also supports an in-house sales and marketing team. Expansion adds general sales and marketing workflows to the same record model; it does not create separate GTM, design and CRM products.

Role	Primary job	Required access
Agency/workspace owner	Configure clients, services, users and integrations	Authorized client workspaces and administrative settings
Campaign operator	Research, segment, draft and prepare outreach	Assigned accounts, campaigns, tasks and channel preparation
Sales representative	Qualify leads, manage opportunities and follow up	Assigned contacts, deals, communications and activities
Marketing lead	Plan campaigns, content, events and reporting	Campaign audiences, briefs, calendars, assets and analytics
Designer/reviewer	Create assets and approve factual/brand claims	Assigned briefs, sources, versions and review decisions
Analyst	Track funnel outcomes and report to clients	Permitted reporting data; no automatic send privileges

Role	Primary job	Required access
Client viewer	Review agreed deliverables and results	Explicitly shared records/assets; broader portal in a later phase

Buyer demand, willingness to pay and performance improvement remain hypotheses. Customer acquisition is not a prerequisite to writing or building the phased demonstration.

2. Locked product decisions

1. **English only.** Interface, assistant replies, generated messages, documents, labels, help and acceptance tests use English. No language switch, translated interface or RTL build is planned. Imported source material retains its original text and provenance; it is not silently translated or rejected solely for language.
2. **Twenty is the application foundation.** Reuse its accounts/companies, contacts, opportunities, tasks, notes, views, permissions and available communications capabilities. New product modules sit within its navigation and record experience.
3. **One product design system, derived from Twenty.** Reuse Twenty components, tokens, typography, spacing, interaction states, accessibility and light/dark behavior. Branding is a documented token/component adaptation. Client asset brand kits may differ, but they do not create competing CRM interface systems.
4. **CRM logic can be customized.** Use supported objects, fields, views, workflows, apps, APIs and front components first. Make scoped core changes when a required behavior cannot be achieved through those interfaces. Core changes must have tests, a documented upstream patch and an upgrade strategy; a complete parallel CRM is not the default.
5. **Persistent actionable assistant.** Its launcher stays at the bottom right throughout authenticated application pages. Conversation and work continue across route changes. Collapsing the popup does not discard its context or jobs.
6. **OpenDesign inside the assistant.** A Design mode and expandable preview/editor expose the existing engine. Integrate its actual tools, jobs, templates and artifacts; do not implement a replacement design engine.
7. **All product phases on Azure; Huawei only after qualification.** Build, test and demonstrate P0–P4 on Azure using available credits. The Huawei demo is in December 2026. Migration starts only if the project passes the first section and Huawei credits are awarded and activated; regional service compatibility must also be verified. There is no automatic month-two cloud move. If the project does not pass or credits are unavailable, continue on Azure.
8. **All discussed features are in this PRD.** A feature's phase establishes its delivery order. Inclusion does not mean all phases will be finished in eight weeks. A phase completes only after its acceptance gate passes; unfinished work carries forward visibly.
9. **Commercial actions are real only when reconciled.** A generated document, exported CSV, paused draft, provider submission and confirmed delivery are different states. Fixtures are always identified as simulated.
10. **Source-backed AI and authoritative rules.** AI researches, proposes, drafts, designs and helps act. Permissions, deduplication, exclusions, approved versions, send readiness and reconciliation remain enforced by the application.

3. Reuse and customization architecture

3.1 Twenty

Use a self-hosted Twenty deployment as the system of record and primary web experience. Inventory the selected release before development and record what is native, configurable, an extension, a targeted core change or an external integration. Existing implementations must be tested before their capability is counted as complete.

Published app interfaces cover custom data objects, logic functions, front components and navigation. Front components run in a constrained environment; a persistent global assistant may require a small host-shell change rather than treating a record side panel as a global popup. Validate the integration location in Phase 0. Twenty app interfaces, front-component constraints

Twenty's repository distinguishes AGPL core, packages with separate licenses, commercially marked enterprise files and an Application Exception for qualifying interface-based apps. Modifying AGPL core is a permitted engineering path with its applicable source/license obligations; it is not equivalent to the app exception. Track the exact license of reused files and dependencies, preserve notices, and do not assume enterprise features or trademark rights are granted. Twenty license

3.2 OpenDesign

Run the actual OpenDesign engine behind the assistant's design adapter. Use its existing design-system packages, templates, project/artifact handling and supported revision/export tools. Product-owned work is the authenticated connection, approved campaign brief, client scope, job orchestration, assistant presentation and links back to CRM records.

The integration exposes compact progress and preview in the popup, with an expanded workspace for detailed editing. If the pinned version lacks a supported embeddable editor, present its previews/tools in the assistant and open its existing editor through authenticated project access. Do not promise an iframe integration until it is verified.

The root license is Apache-2.0. Its hosted plugin specification scopes v1 authentication to a single tenant and describes some infrastructure capabilities as planned; confirm shipped behavior at the chosen revision. Use separate client-scoped jobs/projects or isolated engine instances behind the application boundary. Do not rely on a shared daemon token to provide client authorization. OpenDesign license, plugin and hosted-mode specification, design-system conventions

3.3 Product-owned services and data

Component	Responsibility	Reuse boundary
Twenty shell and CRM	Navigation, identity, CRM records, native views and components	Reuse and configure; targeted patches when necessary
Assistant orchestration	Conversation, live page context, tool selection, navigation and action plans	Connect to existing Twenty AI/MCP capabilities where sufficient; add only the missing product behavior
Evidence/knowledge service	Documents, approved services, citations, retrieval and freshness	Product-specific commercial context linked to Twenty

Component	Responsibility	Reuse boundary
Campaign/action service	Versions, readiness, approval, channel jobs and outcome reconciliation	Reuse Twenty workflows/tasks; add durable campaign semantics
OpenDesign adapter	Approved briefs, scoped jobs, previews, revisions and artifact links	Existing OpenDesign engine; no duplicate renderer/editor architecture
Provider adapters	Email, LinkedIn, enrichment, calendars and optional automation bridges	Documented APIs/capabilities with explicit manual fallbacks
Reporting/evaluation	Actual funnel records, model comparisons, audit and export	Reuse available Twenty dashboards; add required calculations

Persist durable business records in Twenty/PostgreSQL. Store documents and assets in object storage; maintain content hashes and metadata links in CRM. Use a compatible queue/cache for durable job orchestration according to the selected Twenty release. Isolate provider credentials from browser/assistant text. SQLite from the earlier demo is not the target production store.

4. Assistant experience and action contract

4.1 Persistent popup and live context

The launcher is visible at the bottom right on authenticated pages, including company, contact, opportunity, campaign, task, asset, report and settings views. The assistant can collapse, expand, resize and open a larger working panel; it must not cover a primary action or make the underlying page unusable. On narrow screens, expansion may use an overlay while preserving a reachable launcher.

“Always viewing” means live application context: current route, workspace, authorized record IDs, selection, filters, unsaved-change status and relevant visible content. Subscribe to route/selection changes. Retrieve additional records through permission-checked tools when needed. Show a compact context label, for example “Viewing: Acme / Opportunity”. Do not require the user to reattach a screenshot for normal in-app work. Continuous whole-screen recording, password-field capture and observation of other websites are not required.

The conversation remains available across pages within the current client scope. Switching clients visibly switches conversation/data scope. Pending jobs retain their original workspace and target records; they cannot silently apply to the newly selected client. A stale target or record version requires re-resolution before execution.

4.2 Interaction examples

User request	Assistant behavior	Observable outcome
“What needs attention in this campaign?”	Read the current campaign and explain holds with sources	Links to missing evidence, blocked recipients and sender checks
“Open Noura's company and show the sources.”	Navigate to the authorized company and select the evidence section	Route changes; popup and conversation remain present
“Create a follow-up task for tomorrow.”	Create the task in the current workspace using a typed tool	Task ID, due date, owner and undo/compensation where supported
“Move this deal to Proposal.”	Validate transition and required fields, then update	Before/after stage and activity entry; clear reason if blocked

User request	Assistant behavior	Observable outcome
“Create a one-pager using this client's brand.”	Open Design mode and pass approved CRM data to OpenDesign	Scoped job, progress, preview, source/version references
“Make the title shorter.”	Revise the selected artifact through OpenDesign	New artifact version; approved original remains identifiable
“Prepare outreach for these 50 leads.”	Check exclusions/readiness and prepare a bounded batch	Per-row outcomes; reviewed copy and pending handoffs
“Launch the approved sequence.”	Show the concrete audience, sender, schedule and approved version; enforce authorization	Provider submission followed by reconciliation; no false “sent” status

4.3 Action levels

Action level	Examples	Required behavior
Read/navigation	Search, explain, open a page, select a record	Execute within the user's access; no redundant approval prompt
Reversible single-record change	Create task, update owner/stage, save a draft	Execute a clear authorized request; show result and available undo
Bulk/internal change	Assign a list, merge duplicates, enroll contacts	Show affected scope and exclusions; apply atomically or record each partial outcome; ambiguous merges require review
External commitment	Send messages, launch a campaign, publish content, incur provider spend	Concrete review of scope/version/account/cost where relevant; approval bound to that action, with configured role and budget limits
Destructive/admin change	Delete records, alter permissions, remove integration	Role-specific confirmation, dependency checks and backup/restore or documented irreversibility

Configured, previously approved campaign schedules may execute without prompting for every touch. The approval identifies audience, copy version, channels, sender, schedule and spend limits. Changed scope or expired/invalid approval prevents execution. Generation alone never authorizes sending.

4.4 Tools and execution state

Tools use structured schemas and server-side authorization, not unrestricted browser clicks for primary application actions. Navigation commands use an application route registry. DOM interaction is a capability-limited fallback for an unsupported UI handoff and must report what it actually accomplished.

Each action has workspace, user/role, target IDs, expected record version, request/idempotency key, input/output references and timestamps. Long jobs persist across popup closure and reload. States distinguish proposed, running, awaiting review, awaiting external handoff, completed, failed and cancelled. Partial success returns exact record outcomes and safe retry instructions. An assistant message cannot override a hold enforced by the server.

5. Feature inventory and phase assignments

Phase codes: P0 foundation; P1 GTM; P2 sales; P3 marketing and broader campaigns; P4 advanced operations/integrations; H conditional post-qualification Huawei migration. “P1 → P3” means a limited first release followed by the explicitly described expansion. All rows below belong to the full PRD; none imply existing production readiness.

5.1 CRM, workspaces and the shared design system

ID	Feature and required behavior	Phase	Acceptance
CRM-01	Agency/client workspaces, authorized switcher and separate business data	P0	A user cannot read or mutate an unauthorized client's records, assets, jobs or assistant history
CRM-02	Companies/accounts and professional contacts with relationships	P0	Import attaches a contact to the intended company and preserves source IDs without duplicating native records
CRM-03	Lead ownership, team assignment and routing by documented criteria	P1 → P2	A rule assigns the expected owner; an unassigned/ambiguous lead enters a visible queue
CRM-04	Native opportunity pipelines with configurable stages and fields	P0 → P2	Required-field/stage rules and ownership are enforced in UI, API and assistant tools
CRM-05	Tasks, reminders, notes and shared activity timelines	P0 → P2	Tasks persist, link to records, retain owners/dates and show completed versus overdue
CRM-06	Tags, segments, saved views, filters, search and bulk selection	P1 → P3	Saved views reproduce their criteria; selected record IDs are explicit for bulk actions
CRM-07	Record validation, deduplication and reviewed merge	P1 → P2	Conflicting identifiers cannot silently merge; an approved merge preserves relationships and provenance
CRM-08	Relationship/service history, existing-client status and exclusions	P1 → P2	An existing client or opt-out is excluded from incompatible acquisition campaigns across that workspace
CRM-09	Custom objects/fields and schema changes	P0 → P4	Product fields extend Twenty; migration/backfill tests preserve existing records
CRM-10	Roles, record access, reviewer permissions and client sharing	P0 → P4	Restricted users receive neither hidden fields nor assistant-mediated access to them; shared views have explicit scope
UI-01	Twenty-derived tokens/components and documented product branding	P0	A component/token inventory identifies reuse and adaptations; main pages have consistent states and accessible focus
UI-02	English-only desktop/tablet/mobile experience	P0 onward	Every product surface and generated default output is English; no locale switch; primary flows work at target viewports
UI-03	Loading, empty, error, success and partial-job states	P0 onward	Slow requests and failed jobs have actionable feedback; an empty CRM explains how to begin

5.2 AI assistant

ID	Feature and required behavior	Phase	Acceptance
AI-01	Persistent bottom-right launcher, expandable popup and conversation continuity	P0	Launcher survives route changes; collapse/reopen retains the current conversation and job state
AI-02	Live page/selection context and visible context indicator	P0	Selecting another record updates context; hidden or unauthorized fields are absent
AI-03	Search, retrieve, explain and cite workspace knowledge	P1	Factual answers link to the source/record and distinguish a hypothesis or unknown
AI-04	In-app navigation and section selection	P0 → P1	A request opens the expected route/record without losing the assistant session
AI-05	Typed actions for CRM and campaign changes	P1 → P2	A requested action validates permissions/version, creates a durable result and reports its actual state
AI-06	Multi-step plans and bounded batch execution	P1 → P4	User can inspect progress and affected scope, stop pending work and retry only failed steps
AI-07	Proactive next-action suggestions tied to the current page	P1 → P3	Suggestions cite the trigger and offer an action; no unapproved outreach is started
AI-08	Reusable client knowledge, campaign memory and decision history	P1	A settled exclusion/voice/offer correction carries into later drafts for that client
AI-09	Assistant Design mode backed by OpenDesign	P1	A design request invokes a real engine job, shows progress/preview and saves a linked artifact
AI-10	Durable jobs, receipts, cancellation and supported undo	P1 → P4	Closing the popup does not lose a job; cancellation prevents unstarted steps and explains any completed external effects
AI-11	Workspace/user tool policies, approval scopes and spend limits	P0 → P2	Repeated approval is unnecessary within a valid scope; scope/version/budget changes block the action
AI-12	Context changes, stale records and untrusted retrieved instructions	P0 onward	A pending action cannot target a different client's record; retrieved text cannot grant tools or bypass exclusions
AI-13	Model selection, prompt/version records and usage observability	P1 → P4	Each run records model/configuration, sources, latency, cost and outcome without exposing secrets
AI-14	Workflow automation assistance and plain-English rule editing	P4	A proposed rule has a preview/test case; activation is distinct from rule generation
AI-15	Connector-backed analytics explanations and proposed improvements	P3 → P4	Assistant cites property/project, campaign, date range and refresh time; a proposed CRM update or campaign/design revision uses the normal action/approval contract

5.3 GTM research, preparation and targeting

ID	Feature and required behavior	Phase	Acceptance
GTM-01	CSV/XLSX import, field mapping and per-row reconciliation	P1	Every input row has an imported, matched, excluded, rejected or review outcome; retries do not duplicate enrollment
GTM-02	Service/offer catalog, ICP, buyer-role hypotheses and approved proof	P1	Recommendations reference an active offer version and its approved claim scope
GTM-03	Document/playbook upload, parsing and client-scoped retrieval	P1	Supported documents become searchable sources with origin, version and extraction status; a failed parse is visible
GTM-04	Account/person research with dated citations and evidence review	P1	Research records facts, hypotheses and unknowns; a public announcement is not rewritten as an internal-need fact
GTM-05	Account–role–offer ranking with explanations	P1	Ranked offers show role/evidence support; no supporting evidence produces a review hold rather than a fabricated score
GTM-06	Enterprise ABM plans with account briefs and contact-specific rationale	P1	The plan records the recipient, role hypothesis, evidence, offer, sequence and reviewer decision
GTM-07	Professional enrichment with provenance, ambiguity, cache and budget	P1 → P4	Missing data can be requested within a cap; ambiguous results are held; repeated requests reuse valid cached results
GTM-08	English personalized outreach and reusable tone/voice guidelines	P1	Drafts match the role/channel, use only approved claims and retain citations for review
GTM-09	Sequence branches, delays, stop conditions and working schedules	P1 → P3	All touches/branches are reviewable; opt-out/reply stops later touches; schedule/timezone is explicit
GTM-10	Campaign versioning, review decisions and approval invalidation	P1	Changing copy, audience, offer proof or sequence scope invalidates the applicable approval
GTM-11	Deterministic campaign QA, collision checks and exclusions	P1	Duplicates, existing clients, opt-outs, missing fields and overlapping campaigns have visible outcomes
GTM-12	Sender readiness, channel-specific eligibility and domain-review holds	P1	Missing email blocks email without blocking an eligible channel; sensitive sector/offshoring claims require explicit review
GTM-13	Detailed export and visual readiness/prioritization views	P1	Export rows match reviewed records; an export is labeled prepared and does not imply delivery
GTM-14	Partnerships, relationship ownership and campaign coordination	P1 → P3	Partnership contacts have a justified owner; a vendor's HR title is not automatically treated as partnership ownership
GTM-15	Event attendee imports, networking follow-up and assignment	P3	Source/event attribution survives import; existing contacts reconcile; follow-up has owner and due date

ID	Feature and required behavior	Phase	Acceptance
GTM-16	Research refresh, source changes and evidence expiry	P3 → P4	Changed/expired evidence reopens affected review without silently changing approved messages

5.4 Sales and opportunity management

ID	Feature and required behavior	Phase	Acceptance
SALE-01	Lead qualification, disqualification and opportunity conversion	P2	Conversion preserves contact/company/campaign history; disqualification has a reason
SALE-02	Configurable deal stages, ownership, value and expected close date	P2	A stage change obeys configured required fields and records its actor/time
SALE-03	Next steps, activity planning, reminders and overdue follow-up	P2	An active deal without a next step appears in the attention queue
SALE-04	Account/contact interaction history and collision-free handoff	P2	A new owner can see previous approved outreach/replies without duplicating pending touches
SALE-05	Unified authorized communications view and reply triage	P2	Supported replies link to the right contact/campaign/deal; uncertain matches enter review
SALE-06	Meeting preparation, notes, summaries and proposed follow-up	P2 → P4	Summary separates recorded facts from inferred next steps; proposed tasks can be reviewed/applied
SALE-07	Reusable proposals, sales decks, capability statements and case-study assets	P2	An approved offer/proof version populates an editable OpenDesign artifact linked to the opportunity
SALE-08	Basic quotes: scope, line items, amounts and revision/approval	P2 → P4	Totals match inputs; approved revision is identifiable; tax/legal/e-signature integrations are separate capabilities
SALE-09	Won/lost outcomes, reasons and delivery/client handoff	P2	Won/lost updates the funnel; won handoff produces assigned tasks and does not imply service delivery
SALE-10	Pipeline health, weighted forecasts and sales reporting	P2 → P4	Forecast math, stage assumptions and missing values are explicit; AI predictions require their own evaluation
SALE-11	Renewals, expansion opportunities and existing-client account plans	P4	Renewal/upsell is distinguished from acquisition; exclusions are applied according to the campaign purpose
SALE-12	Team playbooks and repeatable discovery/follow-up workflows	P2 → P4	Current playbook version appears in generated preparation; corrections propagate to new work

5.5 Marketing and broader campaign operations

ID	Feature and required behavior	Phase	Acceptance
MKT-01	Campaign briefs: goals, audience, offer, owner, dates and success measures	P1 → P3	Campaign has a measurable objective and named owner before activation
MKT-02	Audience segmentation and saved dynamic/static lists	P3	Segment membership is explainable; approvals bind the intended audience/snapshot policy
MKT-03	Multi-channel campaign coordination and calendar	P3	Touches/content/events share an approved campaign version and do not collide invisibly
MKT-04	Content briefs, English copy generation, revision and review	P3	Copy follows the brief and approved proof; draft, approved and published states are distinct
MKT-05	Email nurture, re-engagement and lifecycle campaigns	P3	Entry/exit conditions, consent/exclusions, schedules and reply stops pass fixture and integration tests
MKT-06	Social content planning and supported publishing handoffs	P3 → P4	Calendar and ready-to-publish assets exist first; unsupported networks receive explicit manual handoffs
MKT-07	Landing-page briefs, design, forms and lead capture	P3 → P4	Captured lead links to campaign/source and deduplicates; a design preview is not labeled a published page
MKT-08	Events/webinars: promotion, attendance and follow-up	P3 → P4	Attendee/import outcomes and attendance status drive reviewed follow-up rather than assumed participation
MKT-09	Partnership/referral campaigns and co-marketing coordination	P3	Partner, client, campaign owner and shared asset approvals are recorded
MKT-10	Paid-campaign planning, creative variants, budget and reporting import	P4	Approved budget/creative plan can be reviewed; imported spend/performance keeps provider/date provenance
MKT-11	Supported paid/social publishing or launch adapters	P4	Provider account/capabilities and explicit launch approval are verified; no adapter is simulated as live
MKT-12	UTM/source attribution, funnel reporting and client summaries	P3	Captured source, campaign, lead, opportunity and revenue links are traceable; unattributed records remain unknown
MKT-13	Controlled experiments and creative/message variants	P4	Variant assignment and exposure are recorded; fixture replies cannot establish commercial lift
MKT-14	Marketing operations dashboard and priorities	P3	Displays current readiness, blocked work, approvals, tasks and real campaign results with refresh timestamps
MKT-15	Campaign measurement plans connecting website behavior to CRM outcomes	P3	Brief defines UTMs, key events, conversion goals and ownership; report distinguishes visits, form submissions, qualified leads, deals and revenue with explicit join/attribution limits

5.6 Design system, OpenDesign and sales assets

ID	Feature and required behavior	Phase	Acceptance
DES-01	Client brand profiles and reusable OpenDesign design-system packages	P1	Brand assets/tokens are versioned; UI design remains Twenty-derived; a client switch selects the correct asset brand
DES-02	Embedded Design mode with job progress, preview and expanded editing	P1	Assistant invokes the actual engine and retains artifact/project IDs across navigation and reload
DES-03	Account research brief	P1	Approved sources, dates, role and offer hypotheses appear in the exported brief
DES-04	Prospect-facing ABM one-pager	P1	Approved copy/proof and client branding populate the artifact; unsupported claims block release
DES-05	Campaign summary and client report	P1 → P3	All chart values reconcile with the campaign snapshot and distinguish simulated events
DES-06	Digital networking card with deterministic QR destination	P1	QR decodes to the stored approved URL; profile links are editable and versioned
DES-07	Proposals, presentations and expanded sales collateral	P2	Artifacts link to the deal, offer scope, source proof and approval version
DES-08	Social/ad/email/landing-page creative templates	P3 → P4	Supported output sizes/formats are capability-checked; variants retain campaign and brand references
DES-09	Conversational revisions, editable source, lineage and approval invalidation	P1 → P4	“Revise this asset” creates a traceable version; changed factual content reopens review
DES-10	Supported exports, responsive previews and asset library	P1 → P3	Export formats match actual engine capabilities; assets remain discoverable and client-scoped
DES-11	Brand/accessibility/content checks and actual-input charts	P1 → P4	Wrong logo/brand/source/chart data causes a visible hold; subjective critique is not presented as objective validation

5.7 Channels, integrations and execution

ID	Feature and required behavior	Phase	Acceptance
INT-01	Smartlead email preparation, sequence/variable mapping and paused drafts	P1	Fixture reconciliation passes; live paused draft requires a verified account and current API contract
INT-02	Approved email launch, delivery/reply events and failure handling	P2	Submitted, accepted, sent, failed and replied are separate reconciled states; retries cannot duplicate sends
INT-03	Dripify personalized CSV/custom fields and capability-aware API adapter	P1 → P4	Supported fields fit limits; unsupported account operations have a visible UI/manual handoff
INT-04	Calls: approved scripts, assigned tasks, outcomes and notes	P1 → P2	Task/script/version/owner are linked; a recorded task is not a completed call

ID	Feature and required behavior	Phase	Acceptance
INT-05	WhatsApp prepared copy/manual tasks; authorized business API expansion	P1 → P4	MVP is an explicit manual handoff; later automation requires supported account/templates/policies
INT-06	Email and calendar synchronization through native Twenty capabilities	P2	Supported account consent, sync and record linkage pass tests against the selected release
INT-07	Professional data/enrichment providers	P1 → P4	Results include provider/time/provenance, confidence/ambiguity and budget usage
INT-08	External CRM connectors: HubSpot, Salesforce, Odoo and other supported CRMs; mapped imports, then selective API/webhook sync	P1 → P4	Imports reconcile first; each live connector declares supported objects, read/write directions, plan/API dependencies and system-of-record/conflict/deletion policies
INT-09	Make/webhooks as optional integration bridges	P4	Scenario creation/activation are distinct; bridge retries preserve action keys and client scope
INT-10	Forms, events and supported social/ad platform connectors	P3 → P4	Each connector has a tested capability matrix and ownership/credential mapping
INT-11	Provider-event inbox, reconciliation ledger and visible handoffs	P1 onward	Duplicate/out-of-order events do not create duplicate actions or regress terminal state
INT-12	Credential health, rate limits, reconnect and connector status	P1 → P4	Expired credentials pause affected jobs; other eligible channels remain usable
INT-13	Connector catalog and per-client connection setup	P1 → P4	Admin sees implemented/planned/manual capabilities, account/property/project, granted scopes, field mappings, refresh schedule, last success/error and disconnect controls; no universal connector claim
INT-14	GA4 read-only reporting connector	P3	Authorized property reports campaign/source/medium, traffic, engagement and supported key-event metrics with date range/timezone/freshness; totals reconcile with equivalent property queries and known reporting limits
INT-15	Microsoft Clarity dashboard-data connector and native-view handoffs	P3	Authorized project imports supported engagement/friction metrics with source/campaign/page dimensions; quota/window/truncation are visible; heatmaps/recordings open authorized native views where available
INT-16	Cross-source identifiers, attribution mappings and metric normalization	P3 → P4	Campaign/UTM/property/project/CRM IDs are mapped explicitly; replayed imports do not double-count; unmatched records remain unknown; anonymous sessions are not automatically identified as CRM contacts

5.8 Automation, administration, reporting and AI evaluation

ID	Feature and required behavior	Phase	Acceptance
OPS-01	Versioned rules/workflows for routing, readiness and follow-up	P1 → P4	Each rule has test inputs/expected outcomes; activation and rollback are recorded
OPS-02	Review queues, approval delegation and notifications	P1 → P3	Assigned reviewer sees pending work; reminders do not count as completed approval
OPS-03	Durable jobs, schedules, retries, idempotency and reconciliation	P1	Worker restart/retry/duplicate event produces one intended business effect
OPS-04	Audit log and assistant action receipts	P0 onward	Actor, client, record version, approval and outcome reconstruct the change
OPS-05	User/team settings, integrations, client brands and offer administration	P0 → P2	Settings changes have role checks and appropriate downstream invalidation
OPS-06	Usage, provider/model budgets, quotas and cost reporting	P1 → P4	Jobs obey workspace caps; estimated and actual costs are distinct
OPS-07	Backups, restore, retention, export and scoped deletion	P0 → P4 → H	Restore test reproduces records/assets; deletion handles linked assets and prevents unwanted reimport
OPS-08	Client reporting/sharing; expanded client review portal	P3 → P4	Shared output exposes only approved scope; portal access is revocable and tested
OPS-09	Operational/model monitoring and incident recovery	P0 onward	Failures can be traced by job/correlation ID without exposing provider secrets
EVAL-01	Account–role–offer ranking experiment	P1 → P4	Rules baseline is frozen; independent held-out relevance results exist before an improvement claim
EVAL-02	Retrieval and grounded-generation evaluation	P1 → P3	Recall@5, factual support, role fit and reviewer edit effort use independent labels and source IDs
EVAL-03	Assistant navigation/action correctness and adversarial tests	P0 onward	Wrong-client, stale-record, ambiguous request and retrieved-instruction cases cannot bypass application policy
EVAL-04	Model adaptation/fine-tuning where useful	P4	Compare frozen embeddings/reranker and bounded adaptation on company-held-out data; keep the stronger baseline
EVAL-05	Commercial-performance experiments once real outcomes exist	P4	Response/conversion claims use identified real cohorts and limitations, never synthetic provider fixtures

5.9 How external connectors support GTM and campaigns

External systems are part of the product's operating model. Twenty remains the local CRM foundation; customers may also keep an existing CRM as the authority for selected objects/fields. Connector-specific mappings define which system owns each field, permitted write directions and conflict resolution. Start with imports/read-only access; add reviewed write-back and continuous sync per tested connector. HubSpot is one concrete example of APIs for CRM objects and webhooks for synchronization. Salesforce/Odoo and further vendors remain individually capability-verified targets. HubSpot integration APIs, webhook configuration

Connected source	Role in GTM/campaign workflow	Product use
External CRM	Customer/account context, ownership, existing opportunities and downstream sales outcomes	Targeting exclusions, qualification, handoff, selective sync and campaign-to-deal reporting
GA4	Website/app traffic, engagement and supported event/conversion reporting	Campaign/source/medium reports, landing-page performance and acquisition analysis
Microsoft Clarity	Behavior and friction on instrumented pages	Identify poor engagement or click friction, review native heatmaps/recordings and propose asset/page improvements
Ad, social, email, form and event platforms	Channel execution, spend, delivery and captured leads	Coordinate approved campaigns and reconcile actual outcomes through the corresponding adapters

GA4's Data API provides programmatic report access; campaign conversion reports are documented. Property configuration, compatible metrics/dimensions, quota and reporting/attribution settings constrain results. Reporting access does not install website tracking or configure key events: include a separate verified instrumentation checklist for customer pages. GA4 Data API, campaign conversion reporting

Clarity's dashboard export API currently allows ten requests per project per day, the previous one to three days, at most three dimensions, and 1,000 rows without pagination. Schedule bounded snapshots, retain observed history, and expose gaps/truncation. Its export does not establish full session-recording/heatmap retrieval or arbitrary historical backfill; use authorized native-view handoffs for those surfaces. Clarity also documents a Google Analytics integration. Clarity export API, Google Analytics integration

Example workflow: launch reviewed outreach with campaign-tagged links → measure website engagement/key events in GA4 → inspect landing-page friction in Clarity → reconcile a captured lead and opportunity in CRM → ask the assistant to explain performance and draft a revised page/asset through OpenDesign. The revision is reviewed before publishing. Correlated campaign metrics are evidence for investigation, not proof of causation or individual identity.

Additional connectors use INT-13's catalog and capability contract; API/account availability, scopes, rate limits and provider plans are verified before delivery. Only authorized client properties/projects are connected. Analytics collection respects configured consent/masking; raw emails, credentials and sensitive CRM fields are not inserted into analytics identifiers or URLs.

6. Core records and consistency rules

Record family	Required information and relationships
Native CRM	Company, person, opportunity, task, note/activity and workspace/user relationships; reuse Twenty IDs
Client commercial knowledge	Offer/version, ICP, buyer-role hypotheses, proof/claim scope, exclusions, playbook, voice and brand profile
Evidence/document	Source URL/file, origin/license where applicable, date, excerpt, extraction version, hash, review decision/reviewer and affected records

Record family	Required information and relationships
Campaign/audience/enrollment	Objective, owner, version, audience snapshot/criteria, contact/source-row IDs, channels, offer, drafts/branches/schedule, exclusions and readiness outcomes
Approval	Reviewer/role, target, copy/asset/audience/version hashes, scope, timestamp, expiry/invalidation reason
Execution/action/event	Idempotency key, provider IDs, actor/client, target/version, approved input, due time, submission/terminal state, event source and simulation flag
Design/project/asset	OpenDesign project/job/artifact IDs, brand/template/brief versions, source CRM records, editable source, export locations, lineage and review state
Assistant/session/context	User/client scope, conversation, current route/selection, tool calls, pending work, context version and durable receipts
Analytics/evaluation	Metric definition, connector/property/project/campaign IDs, actual input references, collection window/timezone/freshness, join policy, provenance, model/dataset revision, labels, split and limitations

Cross-record rules: workspace scope is carried through every query/job/tool; original source rows remain traceable; provider IDs map back to intended CRM records; current approved version is checked at execution; replies/opt-outs cancel applicable future touches; unknown attribution stays unknown. A customer reference cannot substantiate a different service without reviewed proof.

A global company/account exclusion applies to affected contacts in the client workspace, rather than relying solely on an imported per-contact flag. Acquisition, expansion and partnership campaigns use explicit exclusion policies so an existing-client account is treated intentionally.

7. Phases and the two-month plan

Every phase has a demonstrable vertical workflow, a feature checklist, tests and an explicit handoff. Reuse native features before adding development tasks. Parallel development may occur when dependencies permit, but a failed gate does not become complete because the calendar advances.

Phase	Target window/platform	Cumulative scope	Exit demonstration and gate
P0 — Foundation	Week 1 / Azure	Twenty baseline CRM, English-only design-system adaptation, roles/client scope, persistent context-aware assistant shell, navigation, backup/job architecture	Two client workspaces; assistant remains present across native routes; context/navigation/access tests pass; release/capability inventory recorded

Phase	Target window/platform	Cumulative scope	Exit demonstration and gate
P1 — GTM	Weeks 2–3 / Azure	Research/imports, offers/knowledge, evidence ranking, English drafts, reviewed sequences, QA, exports, manual channels, four OpenDesign assets and actionable assistant tools	One supported campaign and one exception campaign complete preparation; real OpenDesign integration works; retries/holds/approvals/source/asset checks pass
P2 — GTM + sales	Week 4 / Azure	Qualification, pipelines, tasks/communications, meetings, proposals/basic quotes, opportunity outcomes and first supported approved email execution	Reviewed lead → opportunity → proposal → follow-up → outcome; assistant acts on CRM; provider fixture suite and any claimed live integration pass
P3 — GTM + sales + marketing	Weeks 5–6 / Azure	Marketing calendar/content, segments/nurture, events/partnerships, asset library, GA4/Clarity connectors, attribution and client reporting; phased form/social capabilities	A cross-channel campaign with content/assets, captured/assigned leads and traceable opportunity/report outcomes; each external capability separately verified

Phase	Target window/platform	Cumulative scope	Exit demonstration and gate
P4 — Expanded operations	Weeks 7–8 / Azure; remaining work tracked	Advanced automations, additional providers/CRM sync, paid/social launch, renewals, portal, experiments, deeper evaluation/adaptation and expanded asset editing	Deliver tested feature groups individually; incomplete IDs remain visible in the roadmap; no blanket “all features done” claim

The delivery goal is to finish and test all product phases P0–P4 on Azure for the December 2026 Huawei demonstration. Native reuse and dependency-based sequencing support that goal; individual feature completion still requires evidence. Track any unfinished feature explicitly rather than declare a phase complete by date. The exact December demo date and competition submission requirements remain to be confirmed.

The demo shows the accepted Azure release: CRM → researched GTM campaign → assistant actions/OpenDesign → sales opportunity → marketing measurement through connected sources, plus an exception/failure path. Present real versus fixture integrations and measured AI results distinctly. Huawei cloud availability is not a prerequisite for the demo or any product phase.

Conditional phase	Trigger/platform	Scope	Exit gate
H — Huawei migration	After the December demo: pass the first section, receive/activate credits, verify compatible services	Migrate the full accepted Azure release, including P3/P4 features, connectors, data, assets and jobs	Full accepted-phase regression and reconciliation suite passes; controlled cutover and rollback rehearsal recorded

If qualification is unsuccessful or credits are delayed/unavailable, keep the complete product operating on Azure. H remains conditional and has no fixed month-two date. Maintain cloud portability during development without consuming the pre-demo schedule on an unapproved destination cutover.

8. Azure through the demo; Huawei after qualification

8.1 Portability from day one

Containerize the pinned Twenty deployment and product workers. Keep PostgreSQL as the authoritative database; avoid changing the business record model during the cloud move. Use storage, queue, model and secret adapters with documented contracts. Provision through versioned infrastructure/configuration and reproducible schema migrations. Separate dev/test from the accepted demo or production-like environment.

These are candidate service mappings, not verified provisioning or interoperability claims:

Capability	Azure development/demo candidate	Conditional Huawei destination	Verification requirement
Application/worker compute	Azure VM/container deployment; managed containers if compatible	ECS/container deployment; managed alternative only when needed	Same images, CPU architecture, network/job behavior and persistent engine workspaces
PostgreSQL	Azure Database for PostgreSQL or managed-compatible deployment	Huawei RDS PostgreSQL or verified equivalent	Versions, required extensions, export/restore, identity and indexing parity
Queue/cache	Redis-compatible service supported by selected Twenty release	Compatible Huawei cache/queue or hosted equivalent	Persistence, TTL, acknowledgements, duplicate handling and worker compatibility
Documents/assets	Azure Blob Storage	Huawei OBS	Provider adapter, hashes, access policy, metadata, signed URLs and OpenDesign storage behavior
Secrets/identity	Azure secret management and application identity	Huawei secret/IAM equivalents	Least-privilege mappings, rotation, credential references and callback URLs
Model inference/evaluation	Azure-hosted or accessible model endpoints/compute using credits where eligible	Verified Huawei inference/ModelArts path or portable model runtime	Model availability, licenses, API adapters, quality/latency/cost and data access
Monitoring/backups	Azure-compatible logs, metrics and backup jobs	Huawei-compatible observability and backup jobs	Correlation IDs, alerts, retention, restore and cost dashboards

8.2 Migration procedure and acceptance

1. Maintain a portability inventory during Azure development: service versions, data volume, models, credentials, connector/property mappings, callback URLs, engine jobs and provider webhook routes. Begin destination provisioning only after passing the first section and receiving/activating Huawei credits; verify region, quotas and compatible services.
2. Deploy the same accepted application/worker images and schema to Huawei staging. Validate storage, database, queue, secrets, network access, OpenDesign and model adapters independently.
3. Copy a consistent database snapshot and documents/assets. Compare record counts, relationships, source IDs and asset hashes; rebuild or transfer indexes as appropriate. Do not treat a database-row count alone as proof of a complete migration.
4. Run the full accepted-phase suite and representative assistant/design/provider workflows on Huawei. If the inference provider/model changes, rerun affected AI quality and tool-contract checks; changing clouds does not establish model equivalence.

5. Set a bounded cutover window. Drain/pause source workers, freeze writes briefly or use a tested replication strategy, capture the final delta and switch application routes/callbacks. Only one environment may execute each live campaign schedule.
6. Reconcile queued/running jobs, approvals, timestamps, provider events and emails/replies across the boundary. Duplicate incoming events must remain idempotent. Communicate a tested interruption window before a live cutover.
7. Validate on the destination and rehearse rollback. Keep the Azure source recoverable for an agreed period; after destination writes occur, rollback must reconcile that delta rather than simply restoring a stale source snapshot.
8. Decommission Azure resources only after destination acceptance and explicit operational authorization. Record final monthly cost, retained backups and migration outcome.

Target migration checks: zero unexplained record/asset loss; unchanged client access boundaries; no duplicate outbound action; scheduled jobs and assistant sessions recover as designed; all accepted P0–P4 scenarios and connector contracts pass on the destination. Actual downtime/RPO/RTO targets are set from Phase 0 measurements and recorded before cutover; none are assumed from a cloud brand.

9. AI contribution and measurement

AI is a substantive product capability from P1: retrieval over client knowledge, supported offer ranking, English drafting, context-aware navigation/action planning, design orchestration and next-action suggestions. Model providers are replaceable behind the tool/retrieval contracts. Do not confuse a chatbot panel with completed action capability, or deterministic template smoke tests with a neural quality evaluation.

For the account–role–offer experiment, retain the proposed 80 company packs, three offers and two roles: 480 relevance judgments. Two independent reviewers use 0 unsupported, 1 plausible and 2 strongly supported labels and adjudicate disagreements. Split by company/group 56 train / 12 validation / 12 test; duplicated sources and related accounts cannot cross splits. Record reviewer expertise and agreement. Public or synthetic data may demonstrate mechanics; final scores must report dataset provenance and limitations.

Compare transparent rules, frozen multilingual embeddings, a pretrained multilingual reranker and bounded adaptation when data/compute justify it. Multilingual models may support imported source material even though the product itself is English-only. Report NDCG@3, relevant-offer ranking accuracy, latency, cost and errors. Keep the stronger baseline if adaptation fails on held-out data. Scores do not represent reply or purchase probabilities. A concrete candidate is BAAI's multilingual reranker; exact revision/license/runtime must be recorded before use.

Evaluation area	Required measurement and evidence
Retrieval	Recall@5 against independently labeled relevant sources; source/split leakage checks
Grounded generation	Supported factual claims, role/channel fit, English quality and reviewer edit effort
Assistant context/navigation	Correct record/route selection, context refresh, stale-client rejection and task completion
Assistant actions	Tool-schema validity, authorization, action correctness, safe retries, approvals and audit receipts

Evaluation area	Required measurement and evidence
OpenDesign integration	Correct brief/brand/version, successful editable output, chart/QR correctness and reviewer acceptance
Cloud migration	Workflow parity, reconciled durable data/jobs, latency/cost and model parity when changed
Commercial outcomes	Actual delivery/reply/meeting/opportunity results with attribution limits; fixture events excluded

Set numerical release thresholds after the first frozen baseline and before the affected test run. Permission bypass, wrong-client mutation, duplicate send and unsupported factual release are blocking failures regardless of average quality. Every completed phase reports its measured results and unconfirmed capabilities honestly.

10. Acceptance scenarios and phase test gates

Scenario	Required result	Earliest gate
English-only product	No language selector/localized surface; all assistant/default generated outputs are English	P0/P1
Assistant persistence	Navigate company → contact → campaign → opportunity; conversation and launcher persist	P0
Live context and navigation	Select a new record; assistant identifies it and can open its related page	P0/P1
Client boundary	Switch clients while a job is pending; job remains pinned to its original authorized client	P0
Native reuse	Main flows use Twenty records/components; exceptions/core patches are listed and tested	P0 onward
Lead import replay	Duplicate Foodics-style rows and repeated imports reconcile; every source row has an outcome	P1
Exclusions/readiness	Existing clients/opt-outs remain excluded; missing email holds email while eligible channels remain available	P1
Role/evidence mismatch	A title or company announcement cannot substantiate an unsupported internal need/offer	P1
Copy/audience/asset change	Applicable approval is invalidated; stale scheduled work cannot execute	P1
Real OpenDesign path	Assistant creates/revises an artifact through the pinned engine; preview/version/editable output are linked	P1

Scenario	Required result	Earliest gate
Asset correctness	Brief citations, actual chart values and QR destination match the approved snapshot	P1
Reply/opt-out/retries	Later applicable touches stop; duplicate/out-of-order events do not duplicate actions	P1/P2
Sales action	Assistant qualifies a lead, updates a deal and creates a follow-up task within permission	P2
Provider outcome	Preparation, paused draft, submission and confirmed delivery are distinguishable	P1/P2
Marketing lifecycle	Segment → content → reviewed nurture/event → captured lead → opportunity/report remains traceable	P3
Analytics connectors	Authorized GA4/Clarity queries retain scope/date/freshness, reconcile supported totals, expose quota gaps and cannot identify anonymous visitors by assumption	P3
External CRM sync	Mapped import/sync preserves IDs, respects field ownership and exclusions, rejects conflicting updates and does not loop or duplicate records	P1/P4
December demonstration	Accepted P0–P4 workflows run on Azure; real/fixture capabilities and any unfinished requirements are explicit	Pre-demo gate
Cloud cutover	Data/assets/approvals/jobs reconcile on Huawei; replayed webhooks cannot trigger duplicate sends	H
AI benchmark	Independent labels, frozen baselines and held-out results accompany any improvement claim	EVAL phase
Failure recovery	Provider outage, source conflict, engine failure and worker restart produce visible, recoverable states	Every affected phase

Definition of done per phase: all assigned first-release feature requirements pass their acceptance criteria; expansions assigned to later phases remain tracked separately; blocked or deferred first-release requirements keep the phase incomplete; local tests and integration checks pass; main UI and assistant flows are reviewed at desktop/tablet/mobile widths; source/permission/version/action checks pass; failure scenarios are exercised; documentation reflects actual capabilities; release commit and known limitations are recorded. After any authorized production deployment, run Playwright against the production URL for the changed flows. A simulated integration cannot satisfy a live capability gate.

Testing is proportional: native unchanged behavior receives targeted integration verification; new rules/actions and data mutations require meaningful tests. Each later phase reruns affected earlier guarantees rather than declaring regression safety from an unrelated historical test count.

11. Operating constraints and implementation guidance

- Reuse before rebuild. A feature's development ticket must state native reuse/configuration, extension, targeted patch or adapter and its evidence. Do not recreate Twenty's data grid, pipeline board, task manager, permission system or component library by default.
- Keep the assistant's tool authority within the user/workspace permissions, including during background jobs. Provider/API secrets never appear in page context, prompts, downloadable reports or logs.
- Approved briefs govern OpenDesign factual content. A generated visual does not create approved commercial proof. Client engine scope and object storage boundaries are tested explicitly.
- Use documented provider APIs for primary execution, with manual handoffs when capability is unavailable. Missing account access is a visible integration dependency, not permission to fabricate a live result.
- Include enrichment/research/model spend caps, timeouts and cancel/retry behavior. No mandatory fine-tuning spend before a comparison makes it useful.
- Consumer advertising, social and paid-channel capabilities enter through the defined P3/P4 modules; the initial evidence remains B2B-heavy. These broader capabilities have separate acceptance and demand hypotheses.
- This PRD does not add unrelated recruitment ATS, payroll, accounting ERP, ecommerce storefront or field-service products. Quote/payment/signature/provider extensions are separate capability decisions when the sales workflow requires them.
- Continue planning/building without requiring customer outreach; preserve a distinction between a working prototype, a supported live integration and evidence of commercial demand.

12. Dependencies, unresolved checks and current-state boundary

Dependency/check	Current knowledge	Required resolution
Product language, reuse and assistant direction	Explicit user decision	Settled in this PRD; no clarification needed
Exact Twenty/OpenDesign release	Prior demo has Twenty SDK source, no native install or real OpenDesign execution	Pin compatible releases; complete native/extension/license capability inventory in P0
Global assistant integration	Required behavior; Twenty front components have a constrained host	P0 spike chooses extension/host-shell patch and tests global context/navigation
OpenDesign inline editor/tools	Engine embedding is desired; exact surface not verified	P0/P1 capability spike; use existing engine/editor with authenticated project access
Azure credits/resources	User reports available credits; all product phases and December demo remain on Azure	Verify account/services/quota and capacity through demo; no balance is asserted here
Huawei credits/region	Conditional on passing the first section after the December demo	Verify award/activation, available services and migration staging before H; remain on Azure otherwise

Dependency/check	Current knowledge	Required resolution
Analytics and external CRM access	APIs exist, but customer accounts/properties/projects are not connected	Verify consent/scopes, mappings, provider limits and current contracts per connector
Models/datasets/reviewers	Prior demo has rules/template code and synthetic unlabeled scaffold	Obtain reviewed evidence and independent labels; run actual baselines
Smartlead/Dripify/other accounts	Prior chats describe a stack, not session access	Verify authorized account capabilities, schema, sender and API limits
Original campaign artifacts	Requested earlier but not available in the implementation record	Audit when available; synthetic demo can proceed with visible origin
Team capacity and phase completion	Two-month horizon provided; throughput not measured	Size tickets after native reuse inventory; preserve complete roadmap with realistic gates

This turn creates the PRD and documentation pointers only. It does not localize or delocalize the earlier demo, implement the popup, fork/install Twenty, integrate OpenDesign, provision/migrate clouds, send outreach or publish content. Prior prototype tests support only their reported local behaviors. They are not acceptance evidence for the new native product requirements.